

BANGOR HARBOUR, United KingdomWMO: **039270**Lat: **54.664N**Lon: **5.668W**Elev: **11**StdP: **101.19**Time Zone: **0.00 (EUW)**Period: **94-07**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.9	1.9	-3.9	2.7	3.4	-2.7	3.0	3.3	20.4	6.1	18.6	6.1	5.8	270	0.733

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	23.2	18.0	21.5	17.0	19.9	16.2	18.5	22.3	17.6	20.7	16.7	19.3	5.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.1	20.4	16.1	11.5	19.3	15.4	11.0	18.3	52.3	22.3	49.4	20.8	46.8	19.5	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.3	12.8	-1.3	26.5	1.3	1.8	-2.2	27.8	-3.0	28.9	-3.7	29.9	-4.6	31.2		
(5)			-1.9	21.2	1.2	1.3	-2.8	22.1	-3.4	22.9	-4.1	23.6	-4.9	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	6.5	6.4	7.4	9.3	11.7	14.2	16.0	16.4	14.5	11.6	8.9	6.9	(6)
(7)		DBStd	4.23	2.16	2.24	2.35	2.40	2.20	2.36	2.19	2.04	1.94	2.19	2.24	2.38	(7)
(8)		HDD10.0	502	111	102	84	41	8	1	0	0	0	10	47	98	(8)
(9)		HDD18.3	2749	368	333	338	272	205	125	78	68	116	208	283	354	(9)
(10)		CDD10.0	812	1	2	5	19	61	127	187	197	136	60	13	3	(10)
(11)		CDD18.3	17	0	0	0	0	0	2	7	7	1	0	0	0	(11)
(12)		CDH23.3	44	0	0	0	0	0	6	21	16	1	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0	(13)
(14)	Wind	WSAvg	6.6	7.7	7.9	7.2	6.6	5.7	5.5	5.0	5.1	5.8	7.3	7.3	7.7	(14)
(15)	Precipitation	PrecAvg	907	89	69	62	62	65	67	70	75	66	95	98	89	(15)
(16)		PrecMax	1221	178	141	125	108	178	197	164	182	156	184	216	178	(16)
(17)		PrecMin	701	22	15	18	14	7	20	18	4	1	27	40	45	(17)
(18)		PrecStd	116	40	34	29	26	37	42	31	41	40	46	43	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.7	14.5	18.6	20.8	23.7	25.6	25.3	22.3	17.4	15.0	14.1	(19)
(20)			MCWB	10.9	11.8	12.0	13.4	15.4	17.6	19.0	19.5	18.0	15.0	13.2	12.7	(20)
(21)		2%	DB	11.7	12.1	13.0	15.5	18.3	21.0	23.5	22.9	20.1	16.2	14.0	12.3	(21)
(22)			MCWB	10.2	10.4	10.8	11.9	13.8	16.0	18.0	18.2	16.6	14.3	12.6	10.9	(22)
(23)		5%	DB	10.8	11.1	11.9	14.0	16.4	19.3	21.4	21.1	18.6	15.5	13.1	11.4	(23)
(24)			MCWB	9.4	9.6	9.9	11.1	12.9	15.4	17.0	17.2	15.7	13.8	11.6	10.0	(24)
(25)		10%	DB	9.9	10.0	11.0	12.7	14.9	17.7	19.5	19.5	17.4	14.9	12.3	10.5	(25)
(26)			MCWB	8.7	8.4	9.3	10.1	12.0	14.5	16.1	16.4	14.9	13.3	10.8	9.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	12.1	12.4	13.9	16.1	18.1	19.8	20.1	18.9	15.6	13.8	12.8	(27)
(28)			MCDWB	12.3	13.5	14.4	17.2	19.7	21.7	24.0	24.7	21.5	17.1	14.7	14.0	(28)
(29)		2%	WB	10.5	10.7	11.2	12.6	14.4	16.7	18.4	18.6	17.2	14.8	12.7	11.2	(29)
(30)			MCDWB	11.5	11.8	12.6	15.2	17.6	20.4	22.9	22.1	19.6	15.9	13.7	12.1	(30)
(31)		5%	WB	9.7	9.7	10.2	11.4	13.3	15.6	17.3	17.7	16.1	14.2	11.9	10.3	(31)
(32)			MCDWB	10.6	10.9	11.6	13.6	15.8	18.6	20.8	20.4	17.9	15.3	12.9	11.2	(32)
(33)		10%	WB	8.9	8.6	9.4	10.3	12.3	14.8	16.4	16.8	15.3	13.6	11.0	9.4	(33)
(34)			MCDWB	9.8	9.8	10.8	12.2	14.5	17.4	19.1	19.0	16.9	14.8	12.2	10.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.4	3.9	4.2	4.9	5.3	5.5	5.4	4.8	3.8	3.7	3.5	(35)	
(36)			MCDBR	4.6	4.9	5.5	7.0	8.0	8.2	8.8	8.2	6.8	4.4	4.5	4.5	(36)
(37)			MCWBR	4.4	4.6	4.3	4.4	4.8	4.8	4.6	4.4	4.4	3.6	4.3	4.5	(37)
(38)		5% WB	MCDWB	4.3	4.7	5.4	6.8	7.3	7.7	8.2	7.4	6.1	4.2	4.2	4.5	(38)
(39)			MCWBR	4.2	4.6	4.5	4.6	4.7	4.8	4.6	4.5	4.5	3.9	4.3	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.347	0.371	0.387	0.391	0.384	0.385	0.370	0.364	0.348	0.328	0.324	(40)	
(41)		taud	2.250	2.330	2.304	2.300	2.338	2.390	2.391	2.448	2.457	2.431	2.320	2.215	(41)	
(42)		Ebn at Noon	583	729	796	837	854	863	854	848	803	724	594	503	(42)	
(43)		Edh at Noon	61	78	100	114	116	112	109	98	85	70	56	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.55	1.23	2.40	3.88	4.96	4.99	4.74	3.90	2.72	1.50	0.71	0.41	(44)	
(45)		RadStd	0.06	0.12	0.19	0.30	0.42	0.49	0.40	0.29	0.17	0.16	0.07	0.04	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page